



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Hector Javier Abitia Andrade
27/08/2026

<https://github.com/EpsiDelt/Applied-Data-Science-IBM/tree/main>



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
 - Data Collection
 - Data Wrangling
 - Exploratory Data Analysis with Data Visualization
 - Exploratory Data Analysis with SQL
 - Building an interactive map with Folium
 - Building a Dashboard with plotly Dash
 - Predictive Analysis
- Summary of all results
 - Exploratory Data Analysis Results
 - Interactive Analytics demo in screenshots
 - Predictive Analysis Results

Introduction

Project Background and Context:

In this capstone project, we aim to predict the successful landing of the Falcon 9 first stage. SpaceX advertises rocket launches at a significantly lower cost compared to other providers, largely due to their ability to reuse the first stage of the rocket. By accurately predicting landing success, we can estimate launch costs and provide valuable insights for companies bidding against SpaceX.

Problems We Want to Find Answers To:

- What factors influence the successful landing of the Falcon 9 first stage?
- How can we accurately predict the landing outcome using machine learning models?
- Which machine learning model performs best in predicting the landing success?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - SpaceX APLI and Web Scraping
- Perform data wrangling
 - Web Scraping
- Perform exploratory data analysis (EDA) using visualization and SQL
 - Matplotlib and Seabord, Folium and Dash
- Perform predictive analysis using classification models
 - Logistic Regression, SVM, Decision Tree, KNN

Data Collection

Data collection process involved a combination of API requests from SpaceX REST API and Web Scraping data from a table in SpaceX's Wikipedia entry.

We had to use both of these data collection methods in order to get complete information about the launches for a more detailed analysis.

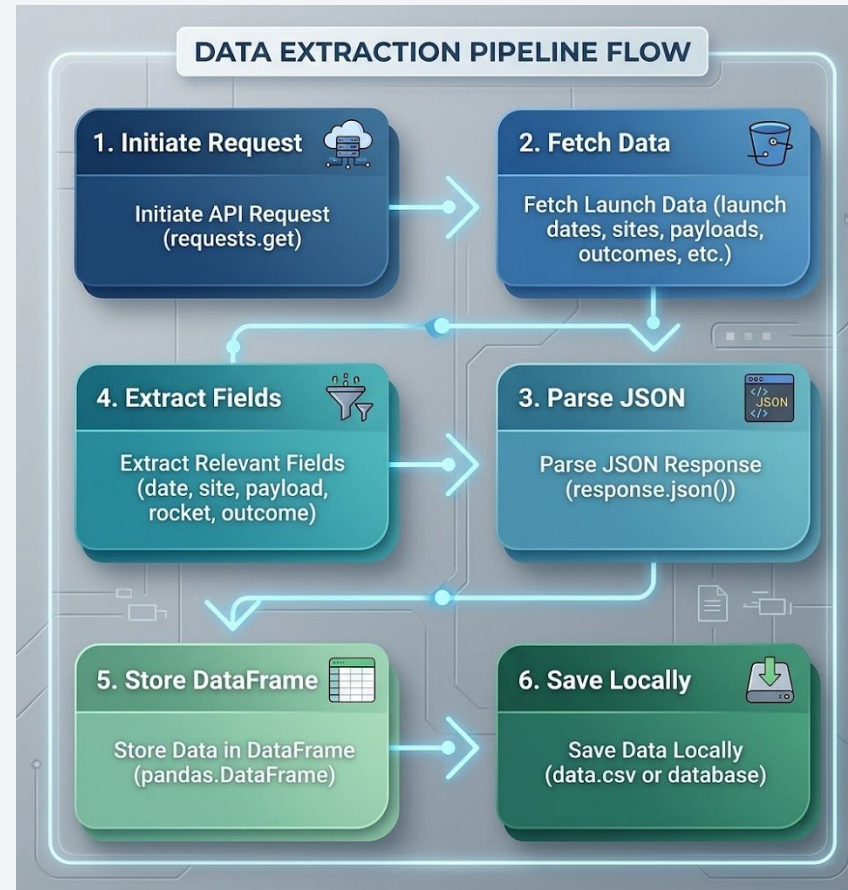
Data Columns are obtained by using SpaceX REST API:

FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, Latitude

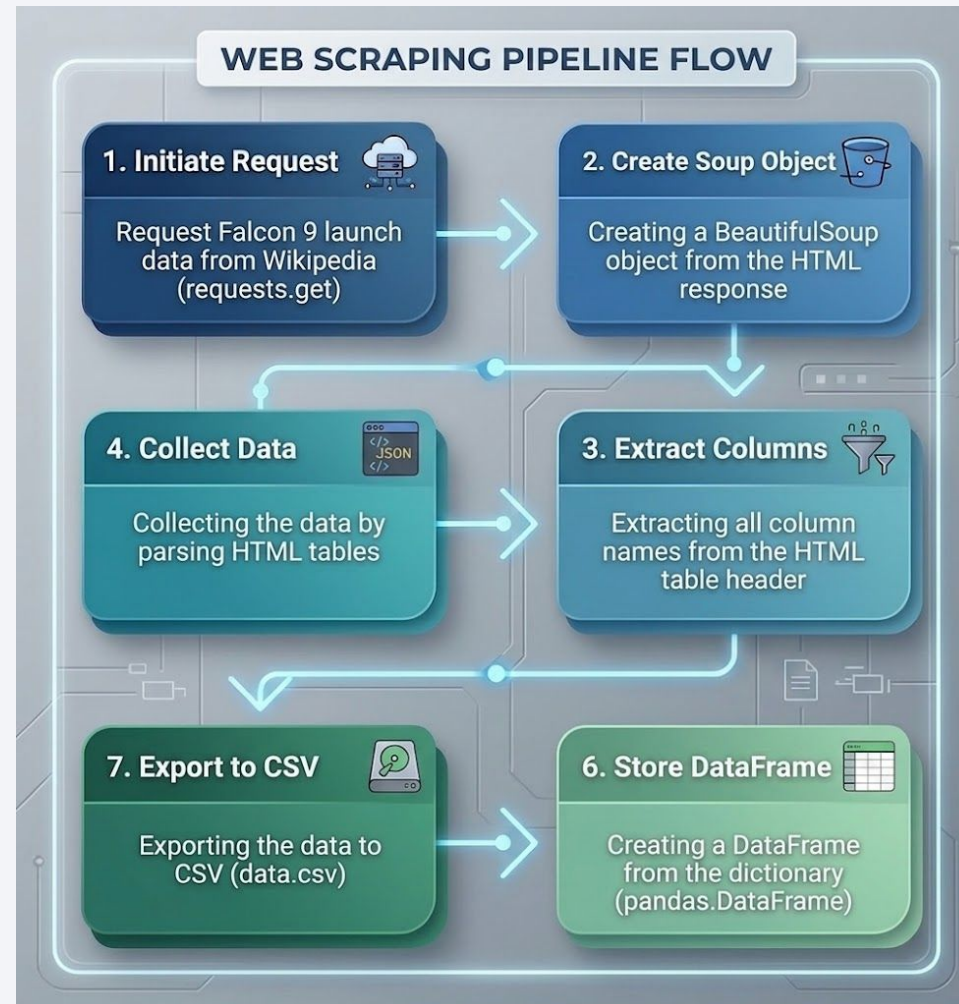
Data Columns are obtained by using Wikipedia Web Scraping:

Flight No., Launch site, Payload, PayloadMass, Orbit, Customer, Launch outcome, Version Booster, Booster landing, Date, Time

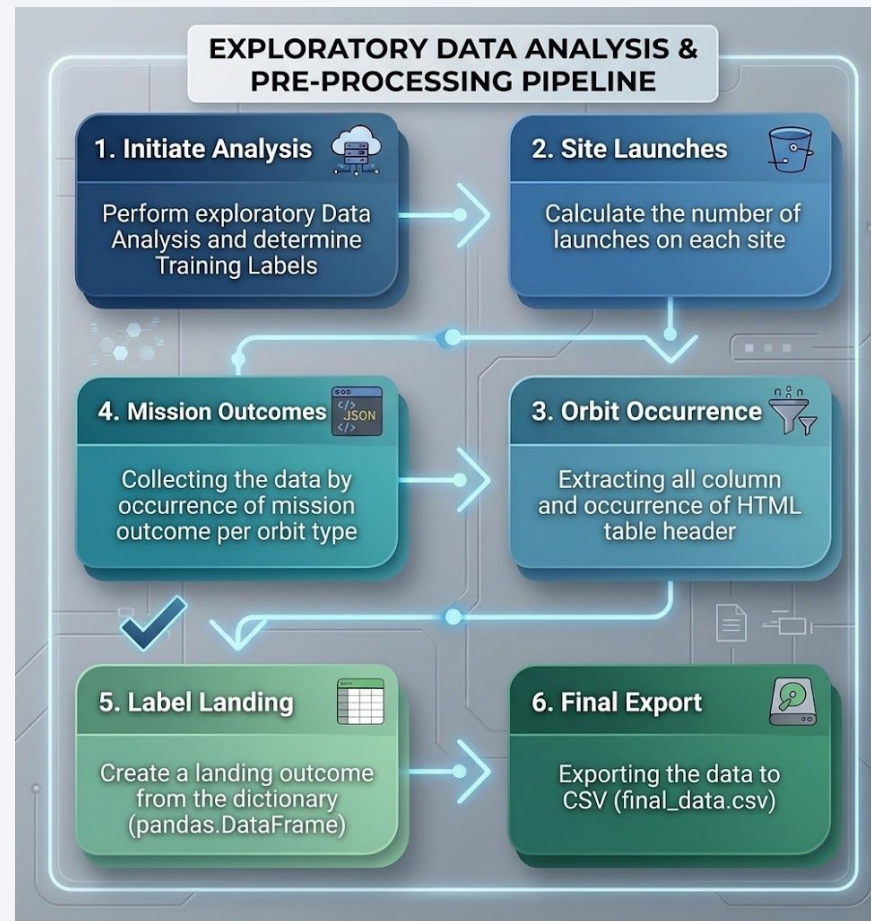
Data Collection – SpaceX API



Data Collection - Scraping



Data Wrangling



EDA with Data Visualization

Data Visualizations & Key Takeaways

Generated Charts:

- Flight Number vs. Payload Mass
- Flight Number vs. Launch Site
- Payload Mass vs. Launch Site
- Orbit Type vs. Success Rate
- Flight Number vs. Orbit Type
- Payload Mass vs. Orbit Type
- Success Rate Yearly Trend

Chart Types & Utility:

- Scatter Plots: Used to evaluate relationships between continuous variables, helping identify feature correlation for machine learning modeling.
- Bar Charts: Highlight differences across categorical groups by comparing distinct categories against a target metric.
- Line Charts: Illustrate time-series trends, tracking performance changes over time.

EDA with SQL

EDA via SQL Operations

Aggregation:

Total launch counts & success/failure distribution.

Success rate calculation segmented by launch site and rocket type.

Data Joining:

Merged primary launch records with auxiliary tables (e.g., rocket specifications).

Integrated datasets for holistic analysis.

Filtering & Sorting:

Filtered data by specific outcomes, timelines, or rocket configurations.

Ordered records by date and performance metrics to identify anomalies.

Advanced Queries:

Applied subqueries to compute derived metrics (e.g., mean payload mass per site).

Build an Interactive Map with Folium

Launch Site Mapping:

- Centered map on NASA Johnson Space Center using precise latitude and longitude coordinates.
- Plotted all global launch sites to evaluate geographical distribution, equator proximity, and coastal access.

Outcome Clustering:

- Implemented MarkerCluster to display launch outcomes per site.
- Color-coded markers (Green = Success, Red = Failure) to visually highlight site performance trends.

Proximity Analysis:

- Drawn dynamic polyline distances from launch site KSC LC-39A to key infrastructure.
- Measured relative distances to critical points: railways, highways, coastlines, and nearest urban areas.

Build a Dashboard with Plotly Dash

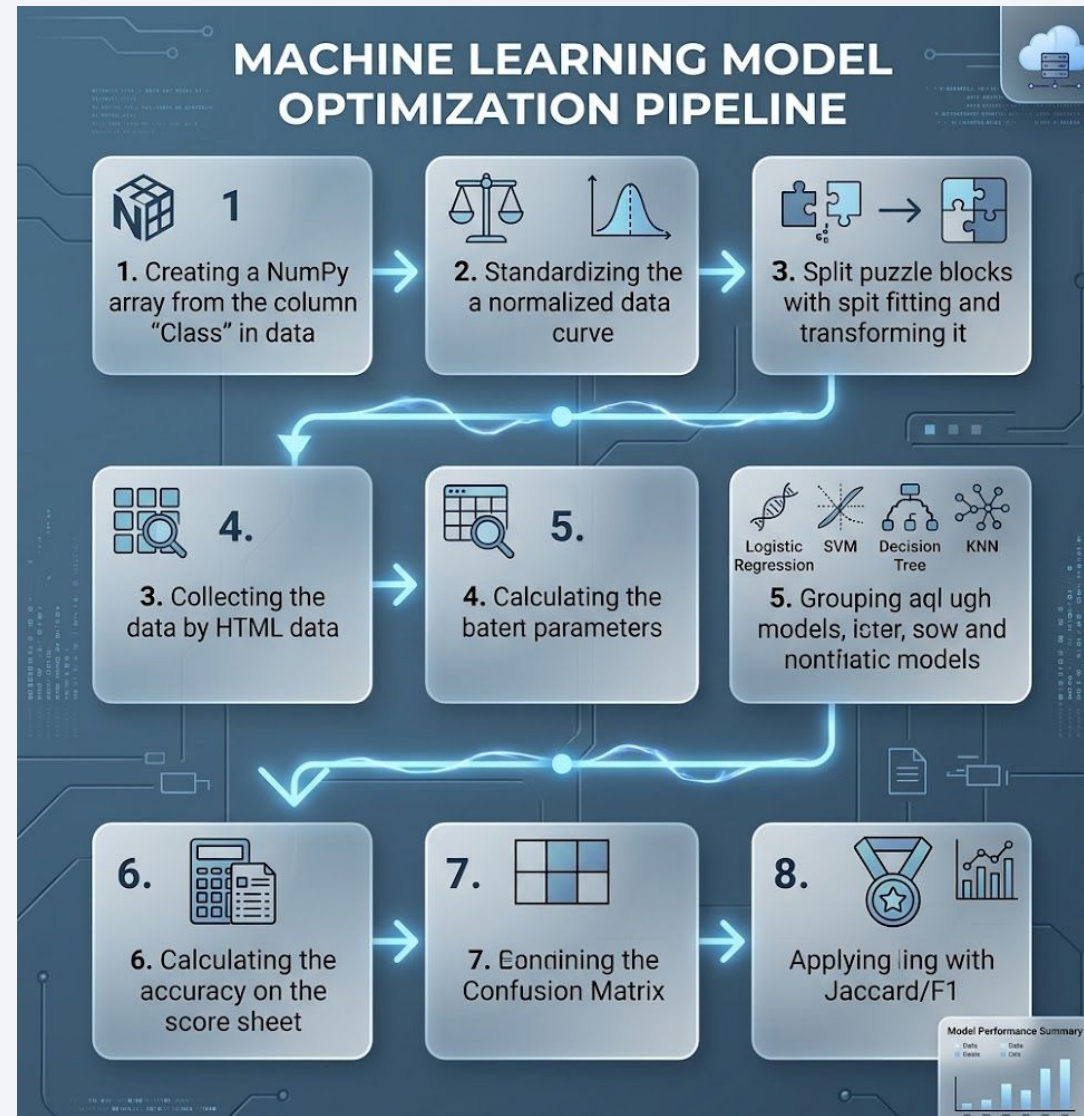
Visualizations Included

- Success Pie Chart: Visualizes success vs. failure distribution across launch sites to evaluate overall performance ratios.
- Payload Scatter Plot: Maps payload mass against launch outcomes to identify optimal weight thresholds for mission success.

Interactive Controls

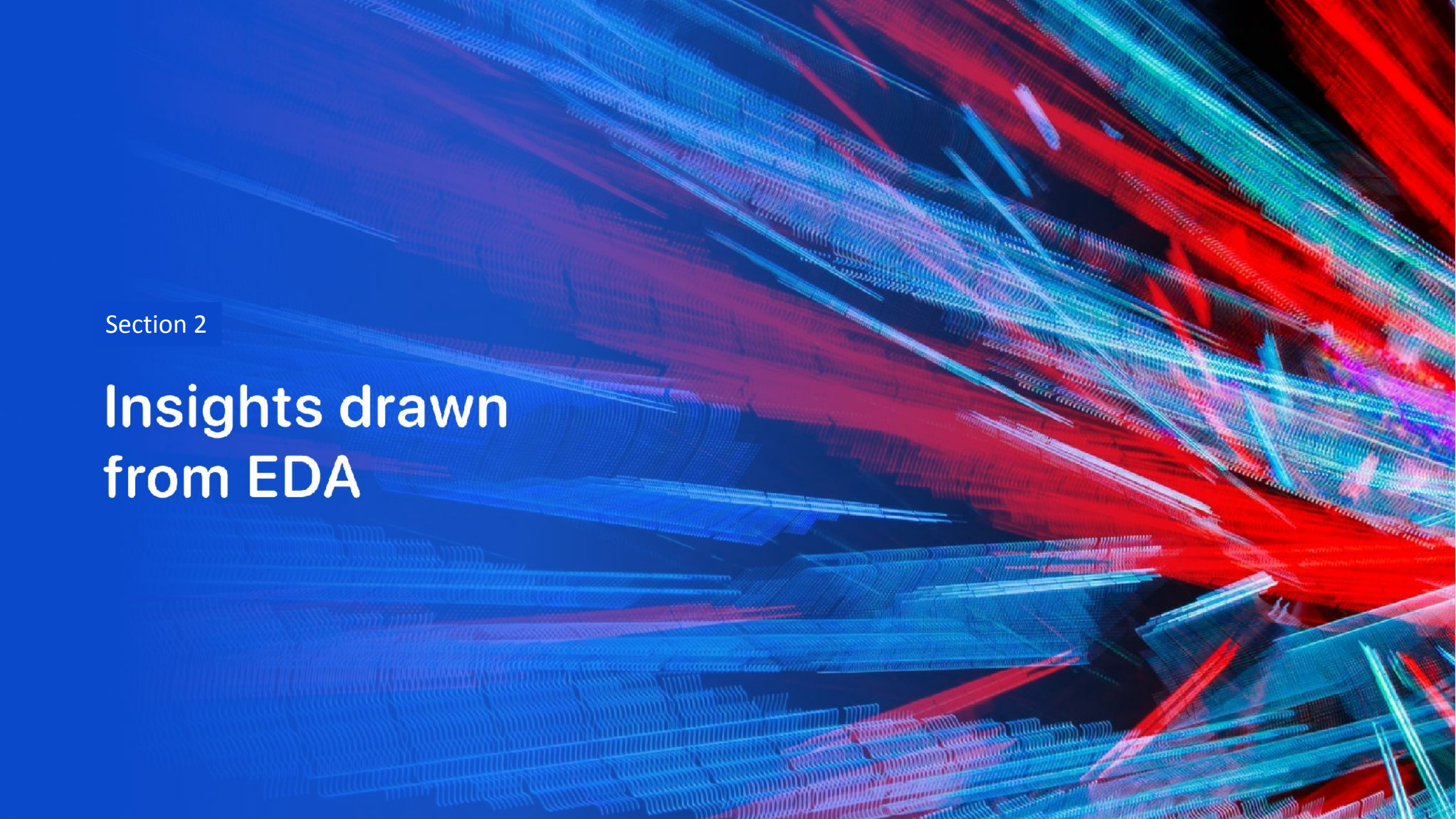
- Site Selection Dropdown: Filters dashboard metrics dynamically by individual launch site or aggregated view.
- Payload Range Slider: Allows real-time filtering across custom payload mass ranges to observe success rate variations.

Predictive Analysis (Classification)



Results

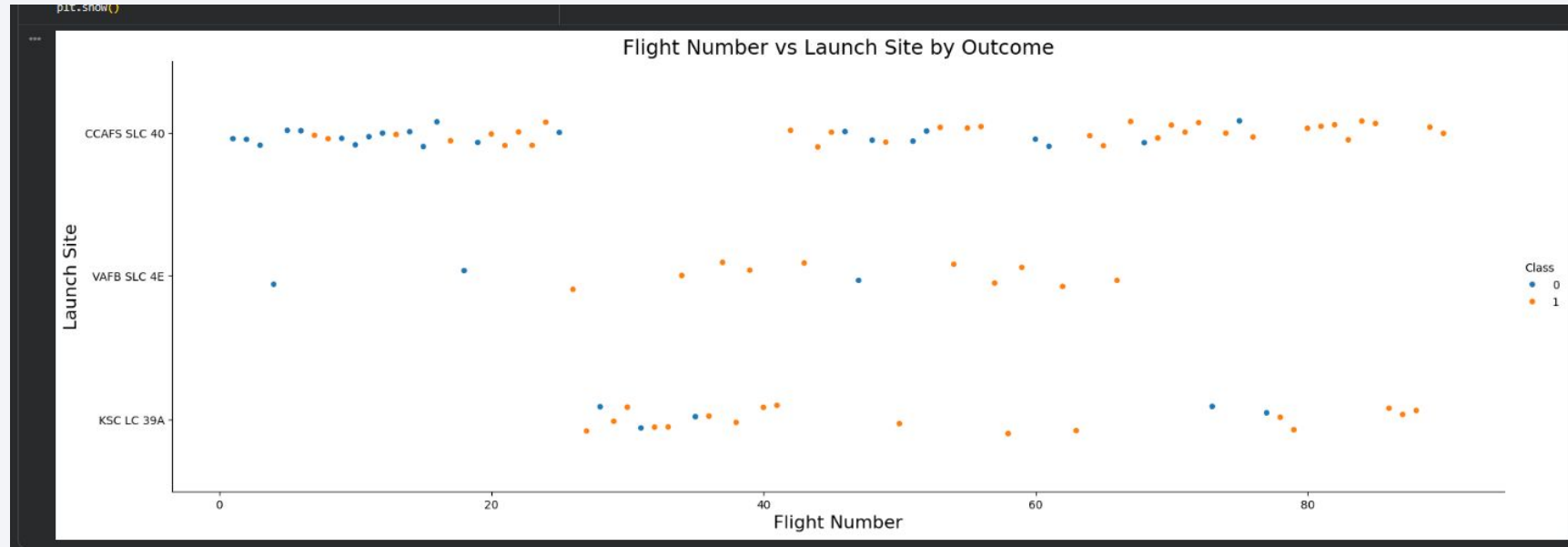
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue and red on the right. These streaks are layered over a fine, light-colored grid, creating a sense of depth and digital complexity.

Section 2

Insights drawn from EDA

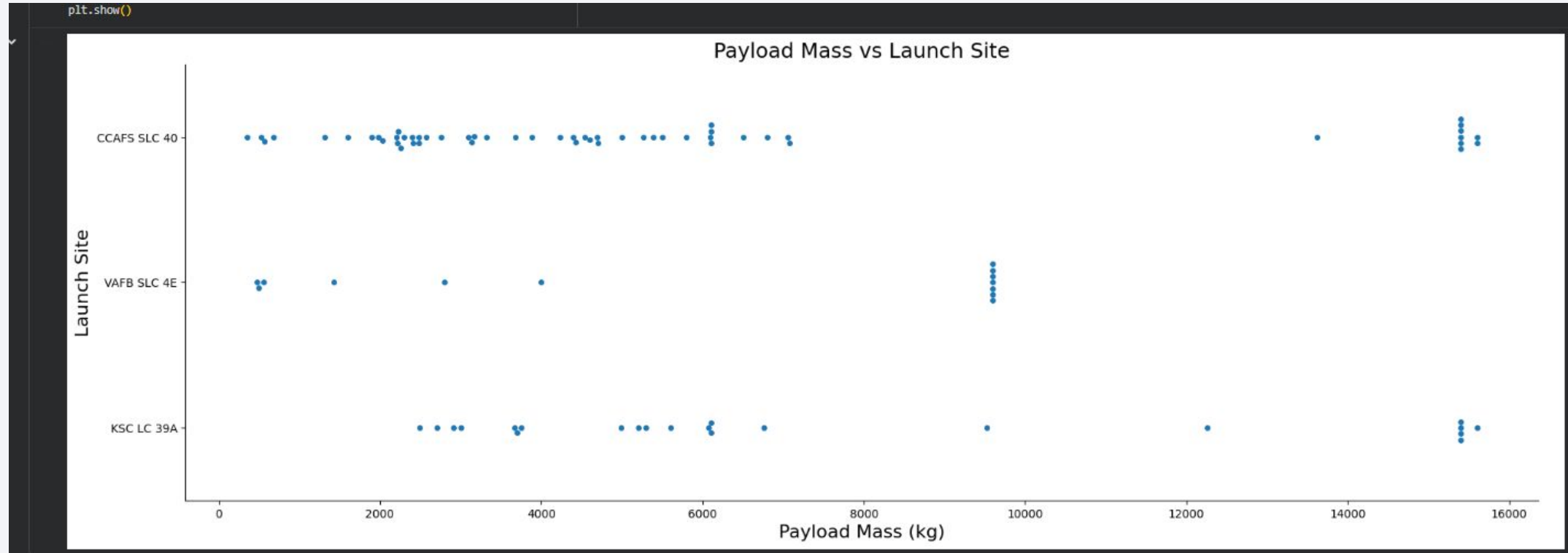
Flight Number vs. Launch Site



Explain:

- **Launch Site Distribution:** CCAFS SLC 40 handled the vast majority of earlier flights (Flight Numbers 1–25) and remains heavily used for later missions. KSC LC 39A and VAFB SLC 4E began handling flights starting around Flight Number 25–30.
- **Success Rate Progression:** Blue dots (Class 0) represent failed mission/landing attempts, while orange dots (Class 1) represent successful ones. Earlier missions (Flight Numbers < 20) show a higher concentration of failures (0). As the flight numbers increase (> 40), orange dots (1) dominate across all launch sites, demonstrating a clear improvement in mission success over time.
- **Site-Specific Performance:** KSC LC 39A and VAFB SLC 4E show a very high proportion of successful outcomes (Class 1), indicating these sites were utilized primarily during a stage of higher operational maturity and reliability.

Payload vs. Launch Site

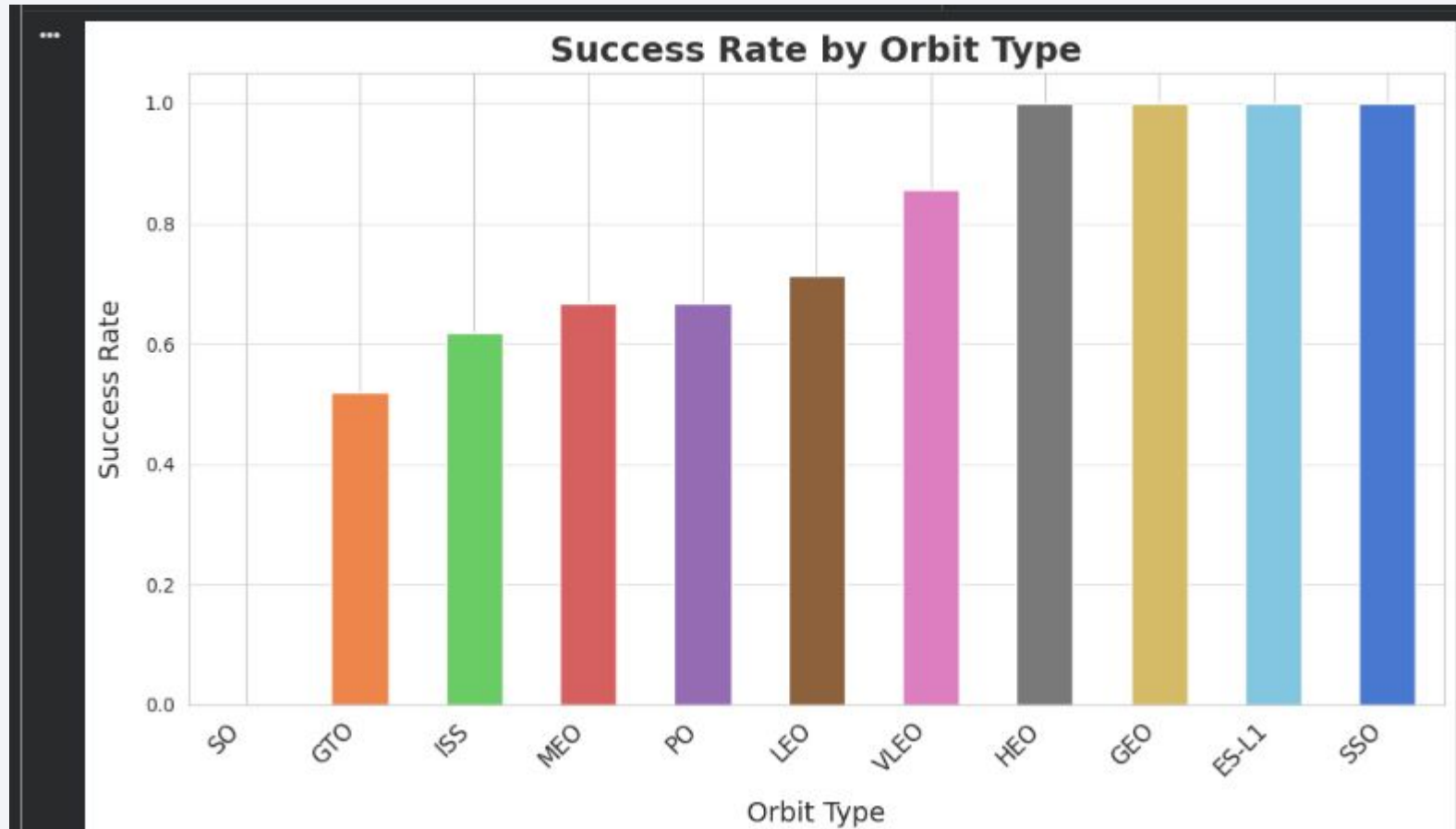


Payload Weight Capability: **CCAFS SLC 40** handles the broadest variety of payload masses, ranging from light loads (around 500 kg) up to heavy payloads near 15,600 kg.

Heavy Payload Clusters: All three sites frequently handle heavy payloads around 15,000–16,000 kg (represented by the dense vertical clusters on the far right). Additionally, **VAFB SLC 4E** shows a distinct vertical cluster near the 9,600 kg mark.

Site Operational Ranges: **KSC LC 39A** generally handles medium to heavy payloads, mostly starting around 2,500 kg and extending to 15,600 kg, with very few light payload launches compared to **CCAFS SLC 40**.

Success Rate vs. Orbit Type

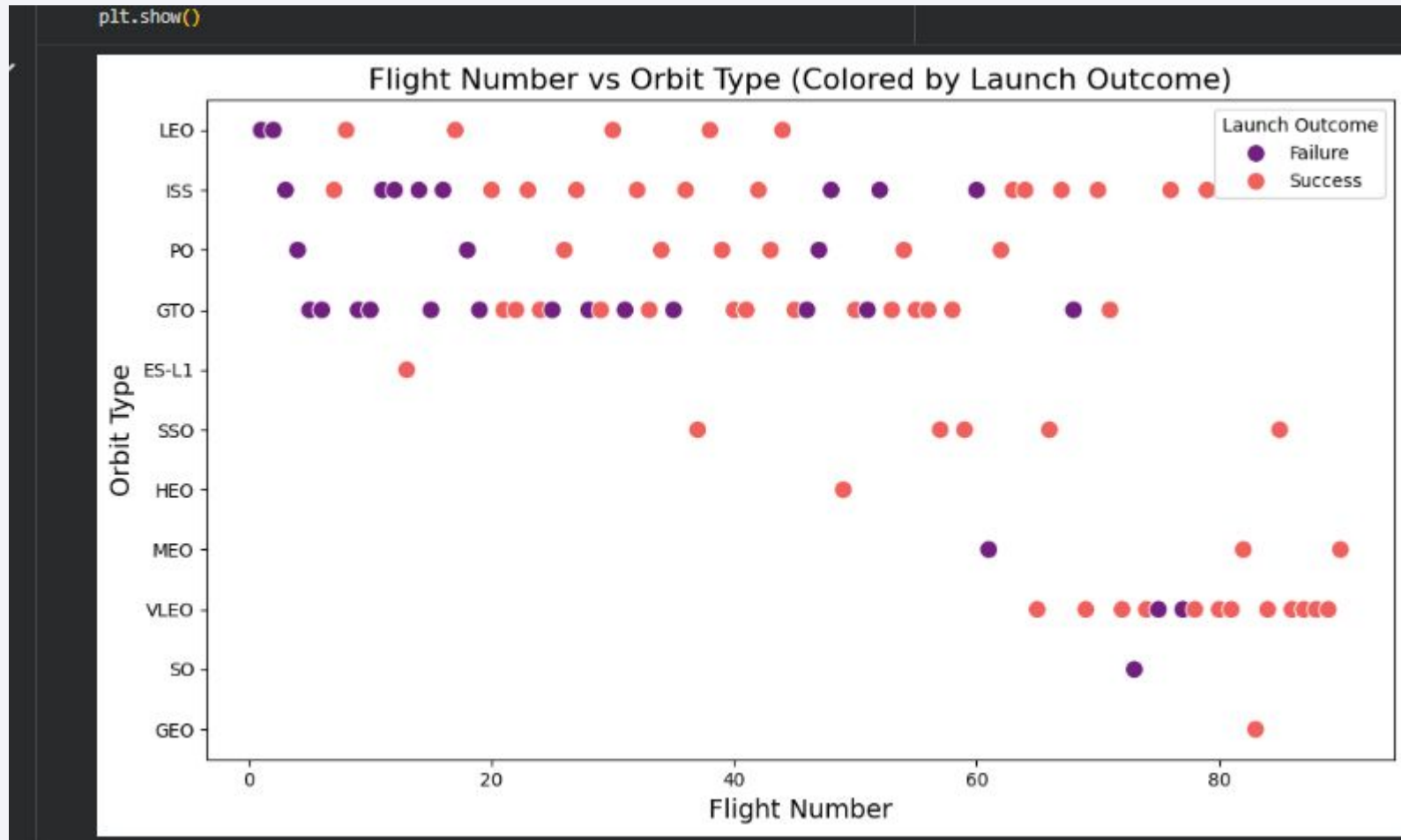


This bar chart illustrates the launch success rate broken down by orbit type.

Key Takeaways

- **Perfect Success Rate (100% / 1.0):** **HEO**, **GEO**, **ES-L1**, and **SSO** orbits all boast a 100% success rate based on the data provided.
- **High Success Rate:** **VLEO** shows a very high success rate of approximately 85% (0.85).
- **Moderate Success Rate:** **LEO** (~70%), **MEO** and **PO** (~66%), and **ISS** (~60%) fall in the moderate performance range between 60% and 72%.
- **Lowest Success Rate:** **GTO** has the lowest success rate among active orbit types at roughly 50% (0.5), while **SO** shows a 0% success rate (no successful launches recorded).

Flight Number vs. Orbit Type

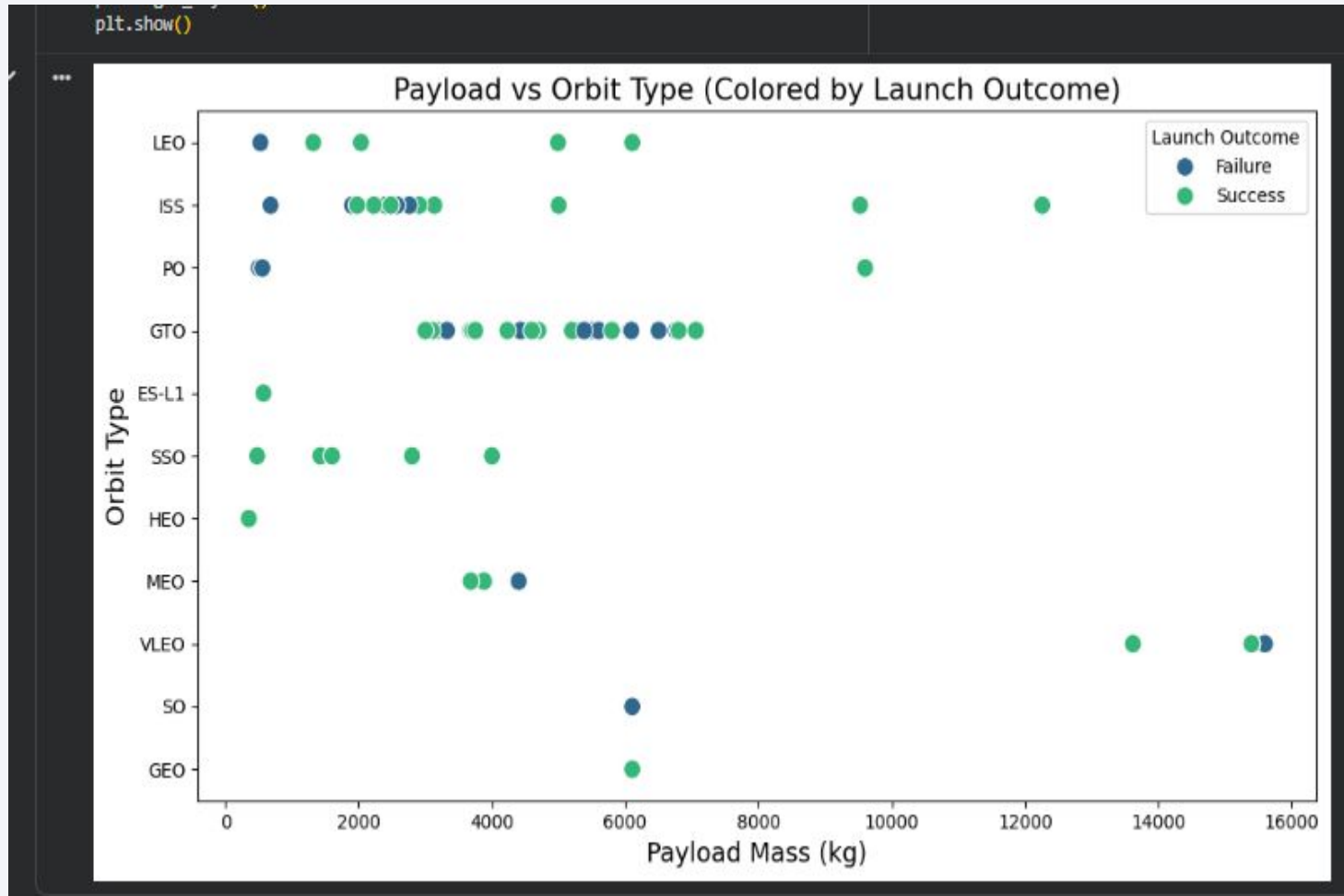


Temporal Shift in Orbits: Early flights (1–40) heavily targeted orbits like **GTO**, **ISS**, **LEO**, and **PO**. Later flights (> 60) show a clear operational shift toward **VLEO** and **MEO**.

Improvement in Success Rate: Purple dots represent failures, which cluster heavily in earlier flights (Flight Numbers < 35) across orbits like **LEO**, **ISS**, and **GTO**. Later missions consist almost entirely of coral dots (successes).

High-Reliability Orbits: Orbits introduced later in the program—such as **VLEO**, **SSO**, **GEO**, and **HEO**—consistently exhibit high success rates with very few failure points.

Payload vs. Orbit Type



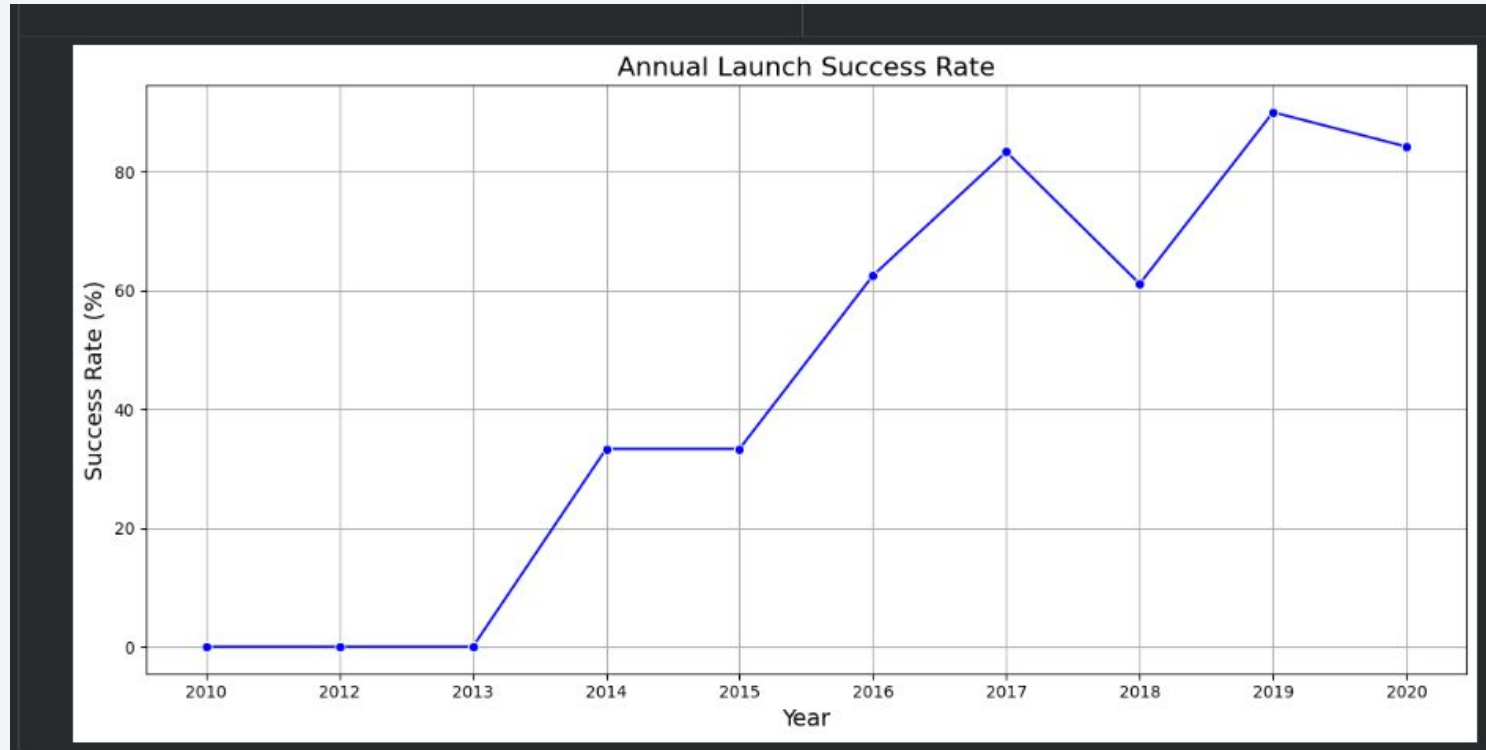
Heavy Payload Concentration: VLEO handles the heaviest payloads (ranging from ~13,500 kg to over 15,500 kg) with a near-perfect success rate. **ISS** and **PO** also record high-weight payloads up to ~10,000–12,000 kg.

Mid-Range Payload Distribution: GTO launches cluster densely in the medium weight class (~3,000 kg to 7,000 kg), displaying a mix of successes and failures.

Light Payload Orbits: Orbits such as **SSO**, **HEO**, and **ES-L1** cater exclusively to lighter payloads (under 5,000 kg) and show consistently successful outcomes.

Failure Clustering: Dark blue dots (**Failure**) are primarily concentrated in lower payload masses (< 1,000 kg for LEO/ISS/PO) or mid-range payloads around **GTO**, while heavier payloads generally achieve higher success rates (**Success**).

Launch Success Yearly Trend



Initial Baseline (2010–2013): The success rate remained at 0% during the earliest years, reflecting the experimental phase of landing attempts.

Steady Growth (2013–2017): A strong upward trajectory began in 2013, rising to around 33% across 2014–2015, reaching ~62% in 2016, and peaking near 83% in 2017.

Program Peak (2019): After a slight dip to ~61% in 2018, the success rate achieved its highest point in 2019 at 90%, before settling at a high ~84% in 2020.

Overall Trend: The graph demonstrates clear technology maturation over time, transitioning from early trial failures to consistently high operational reliability above 80%.

All Launch Site Names

```
%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```


Launch Site Names Begin with 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db  
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Direct Interpretation of SQL Query Results

- **Successful Filtering:** The query correctly pulls the first 5 recorded launches from Cape Canaveral (**CCAFS LC-40**).
- **Early SpaceX Phase (2010–2013):** Captures the historical start of the Falcon 9 rocket program using its earliest iterations (**F9 v1.0 B0003** to **B0007**).
- **Mission Success vs. Landing Experiments:** While 100% of the primary missions successfully delivered payloads to Low Earth Orbit (**LEO / ISS**), rocket reusability was still in its infancy—the first two parachute recovery attempts failed, followed by three missions with no landing attempt.
- **Light Payloads:** Reflects early-stage space station resupply missions for NASA and SpaceX, carrying relatively light test and cargo payloads ranging from 0 to 677 kg.

Total Payload Mass

```
%sql SELECT SUM("PAYLOAD_MASS_KG_") FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
SUM("PAYLOAD_MASS_KG_")
```

```
45596
```

Query Objective: Calculates the total cumulative weight of all cargo launched specifically for NASA's Commercial Resupply Services contract (**WHERE "Customer" = 'NASA (CRS)'**).

Total Payload Mass: The aggregate mass delivered across all NASA CRS missions in the dataset equals **45,596 kg** (approx. 45.6 metric tons).

Core Insight: Highlights the heavy volume of cargo SpaceX transported to the International Space Station (ISS) under its primary resupply partnership with NASA.

Average Payload Mass by F9 v1.1

```
%sql SELECT AVG("PAYLOAD_MASS_KG") FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';

... * sqlite:///my_data1.db
Done.
AVG("PAYLOAD_MASS_KG")
2928.4
```

Query Objective: Calculates the average payload mass launched specifically by the Falcon 9 v1.1 booster variant (WHERE "Booster_Version" = 'F9 v1.1').

Average Payload Mass: The calculated mean payload carried by this booster version is 2,928.4 kg.

Core Insight: Demonstrates the typical medium-weight capacity of SpaceX's second-generation Falcon 9 configuration, which upgraded lifting performance significantly over the original v1.0.

First Successful Ground Landing Date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
%sql SELECT MIN("Date") FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
MIN("Date")
```

```
2015-12-22
```

Query Objective: Identifies the earliest recorded date for a successful booster landing on a ground pad using the `MIN("Date")` function paired with the `'Success (ground pad)'` condition.

Earliest Date Returned: The query returns **2015-12-22** (December 22, 2015).

Core Insight: Marks a major historical milestone in aerospace, capturing SpaceX's very first successful return and vertical landing of a orbital-class Falcon 9 first stage at Landing Zone 1 (LZ-1).

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS_KG_" > 4000 AND "PAYLOAD_MASS_KG_" < 6000;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Booster_Version
```

```
F9 FT B1022
```

```
F9 FT B1026
```

```
F9 FT B1021.2
```

```
F9 FT B1031.2
```

Direct Interpretation of SQL Query Results

- **Query Objective:** Filters for unique booster versions (`SELECT DISTINCT`) that achieved a successful landing on an autonomous spaceport drone ship (`'Success (drone ship)'`) while carrying mid-to-heavy payload masses between 4,000 kg and 6,000 kg.
- **Returned Booster Versions:** The query outputs four distinct Full Thrust (FT) booster designations:
 - F9 FT B1022
 - F9 FT B1026
 - F9 FT B1021.2
 - F9 FT B1031.2
- **Core Insight:** Highlights the capability of SpaceX's Falcon 9 Full Thrust variant to successfully perform offshore sea landings even when carrying substantial mid-weight payloads requiring high energy trajectories.

Total Number of Successful and Failure Mission Outcomes

Task 7

List the total number of successful and failure mission outcomes

```
%sql SELECT "Mission_Outcome", COUNT(*) AS "Total" FROM SPACEXTABLE WHERE "Mission_Outcome" IN ('Success', 'Failure') GROUP BY "Mission_Outcome";
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Mission_Outcome Total
```

```
Success          98
```

Direct Interpretation of SQL Query Results

- **Query Objective:** Groups and counts the total number of records categorized strictly as 'Success' or 'Failure' under the `Mission_Outcome` column.
- **Returned Aggregations:** The output displays a total of **98** successful missions (Success). No rows for 'Failure' appear in the result set.
- **Core Insight:** Demonstrates an overwhelming primary mission success rate across the dataset. The absence of a 'Failure' row indicates that total primary mission failures were either non-existent under that exact string name or excluded due to slight variations in string formatting (e.g., trailing spaces or additional text in the raw data).

Boosters Carried Maximum Payload

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM SPACEXTABLE);
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Booster_Version
```

```
F9 B5 B1048.4
```

```
F9 B5 B1049.4
```

```
F9 B5 B1051.3
```

```
F9 B5 B1056.4
```

```
F9 B5 B1048.5
```

```
F9 B5 B1051.4
```

```
F9 B5 B1049.5
```

```
F9 B5 B1060.2
```

```
F9 B5 B1058.3
```

```
F9 B5 B1051.6
```

```
F9 B5 B1060.3
```

```
F9 B5 B1049.7
```

Query Objective: Uses a subquery with the aggregate function `MAX("PAYLOAD_MASS_KG_")` to identify all unique booster versions (`SELECT DISTINCT`) that carried the maximum payload mass recorded in the dataset (15,600 kg).

Returned Booster Versions: The output lists multiple Falcon 9 Block 5 (F9 B5) boosters, including B1048.4, B1049.4, B1051.3, B1056.4, B1048.5, B1051.4, B1049.5, B1060.2, B1058.3, B1051.6, B1060.3, and B1049.7.

Core Insight: Demonstrates that maximum-weight missions are handled exclusively by the highly reusable **Falcon 9 Block 5** variant, typically associated with heavy Starlink satellite batch deployments.

2015 Launch Records

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)= '2015' for year.

```
%%sql

SELECT
CASE
    WHEN substr("Date", 6, 2) = '01' THEN 'January'
    WHEN substr("Date", 6, 2) = '02' THEN 'February'
    WHEN substr("Date", 6, 2) = '03' THEN 'March'
    WHEN substr("Date", 6, 2) = '04' THEN 'April'
    WHEN substr("Date", 6, 2) = '05' THEN 'May'
    WHEN substr("Date", 6, 2) = '06' THEN 'June'
    WHEN substr("Date", 6, 2) = '07' THEN 'July'
    WHEN substr("Date", 6, 2) = '08' THEN 'August'
    WHEN substr("Date", 6, 2) = '09' THEN 'September'
    WHEN substr("Date", 6, 2) = '10' THEN 'October'
    WHEN substr("Date", 6, 2) = '11' THEN 'November'
    WHEN substr("Date", 6, 2) = '12' THEN 'December'
    ELSE 'Unknown'
END AS "Month_Name",
    "Mission_Outcome",
    "Booster_Version",
    "Launch_Site"
FROM
    SPACEXTABLE
WHERE
    substr("Date", 0, 5) = '2015';
```

```
* sqlite:///my_data1.db
Done.
Month_Name Mission_Outcome Booster_Version Launch_Site
January      Success      F9 v1.1 B1012  CCAFS LC-40
February     Success      F9 v1.1 B1013  CCAFS LC-40
March        Success      F9 v1.1 B1014  CCAFS LC-40
April        Success      F9 v1.1 B1015  CCAFS LC-40
April        Success      F9 v1.1 B1016  CCAFS LC-40
June         Failure (in flight) F9 v1.1 B1018  CCAFS LC-40
December     Success      F9 FT B1019    CCAFS LC-40
```

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
%%sql
SELECT
    "Landing_Outcome",
    COUNT(*) AS "Count"
FROM
    SPACEXTABLE
WHERE
    "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY
    "Landing_Outcome"
ORDER BY
    COUNT(*) DESC;

* sqlite:///my_data1.db
Done.
Landing_Outcome  Count
No attempt      10
Success (drone ship) 5
Failure (drone ship) 5
Success (ground pad) 3
Controlled (ocean)  3
Uncontrolled (ocean) 2
Failure (parachute)  2
Precluded (drone ship) 1
```

Query Objective: Filters for launches in the year 2015 using `substr("Date", 0, 5) = '2015'` and maps numeric month values (`substr("Date", 6, 2)`) to full month names via a `CASE` statement.

Returned Launches (2015): Captures 7 total launches occurring across January, February, March, April (2 launches), June, and December. All missions launched from **CCAFS LC-40** using **F9 v1.1** and **F9 FT** boosters.

Mission Performance: Shows a primary mission success rate of ~86% for 2015, with 6 successful launches and 1 catastrophic in-flight failure in June (**Failure (in flight)**) with booster **B1018**).

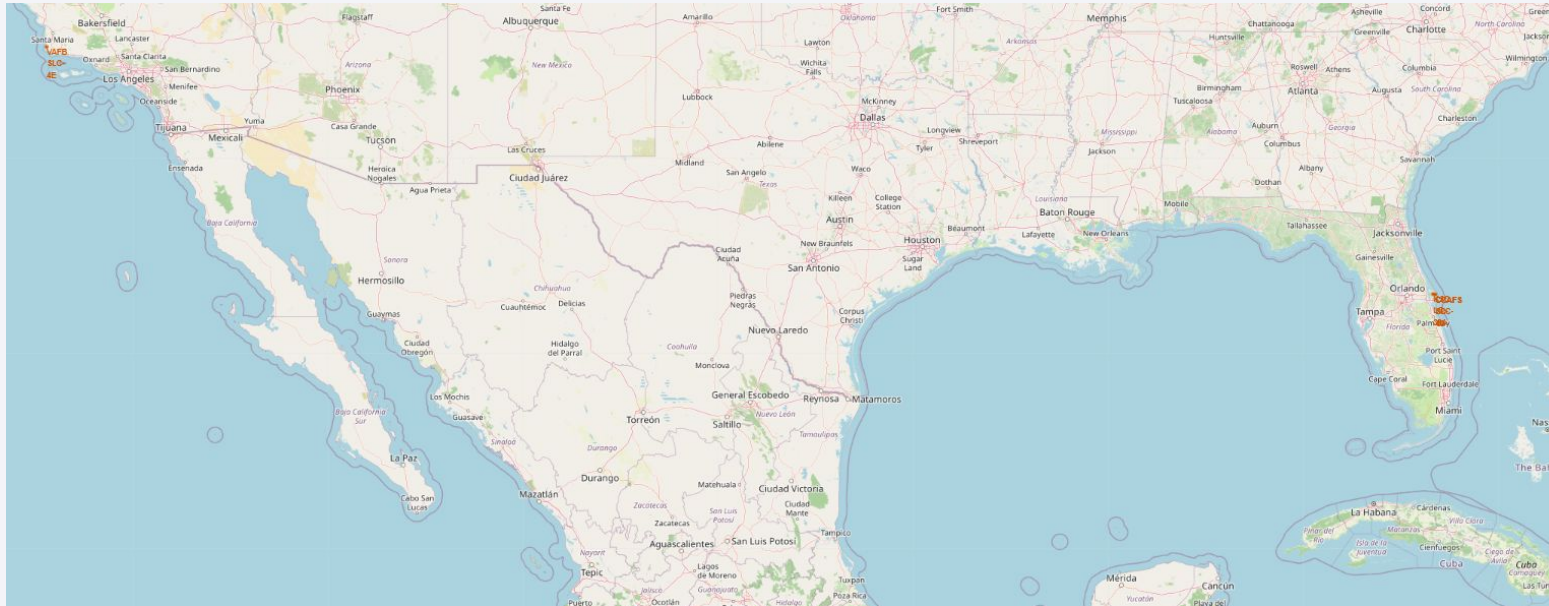
Query Discrepancy Note: The prompt task requested filtering for failed drone ship landings (`Landing_Outcome = 'Failure (drone ship)'`), but the query selected `Mission_Outcome` and omitted the landing outcome condition, thus returning all 2015 launches regardless of landing performance.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is dark blue with a thin white line representing the horizon. The city lights are visible as bright yellow and orange spots against the dark blue background of the night sky.

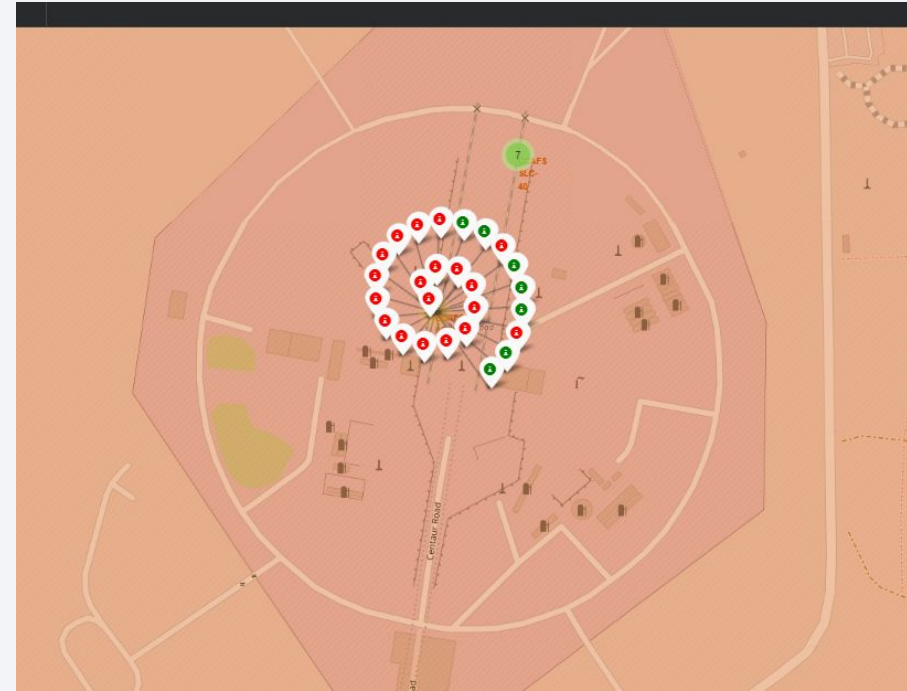
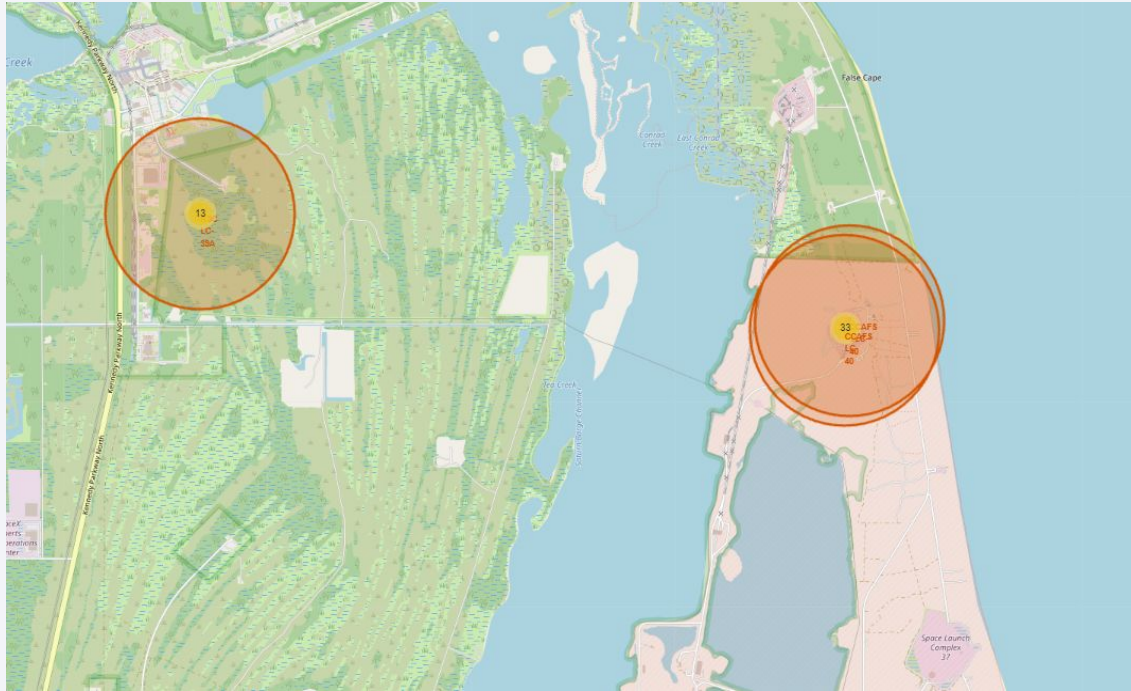
Section 3

Launch Sites Proximities Analysis

<Folium Map Screenshot 1>

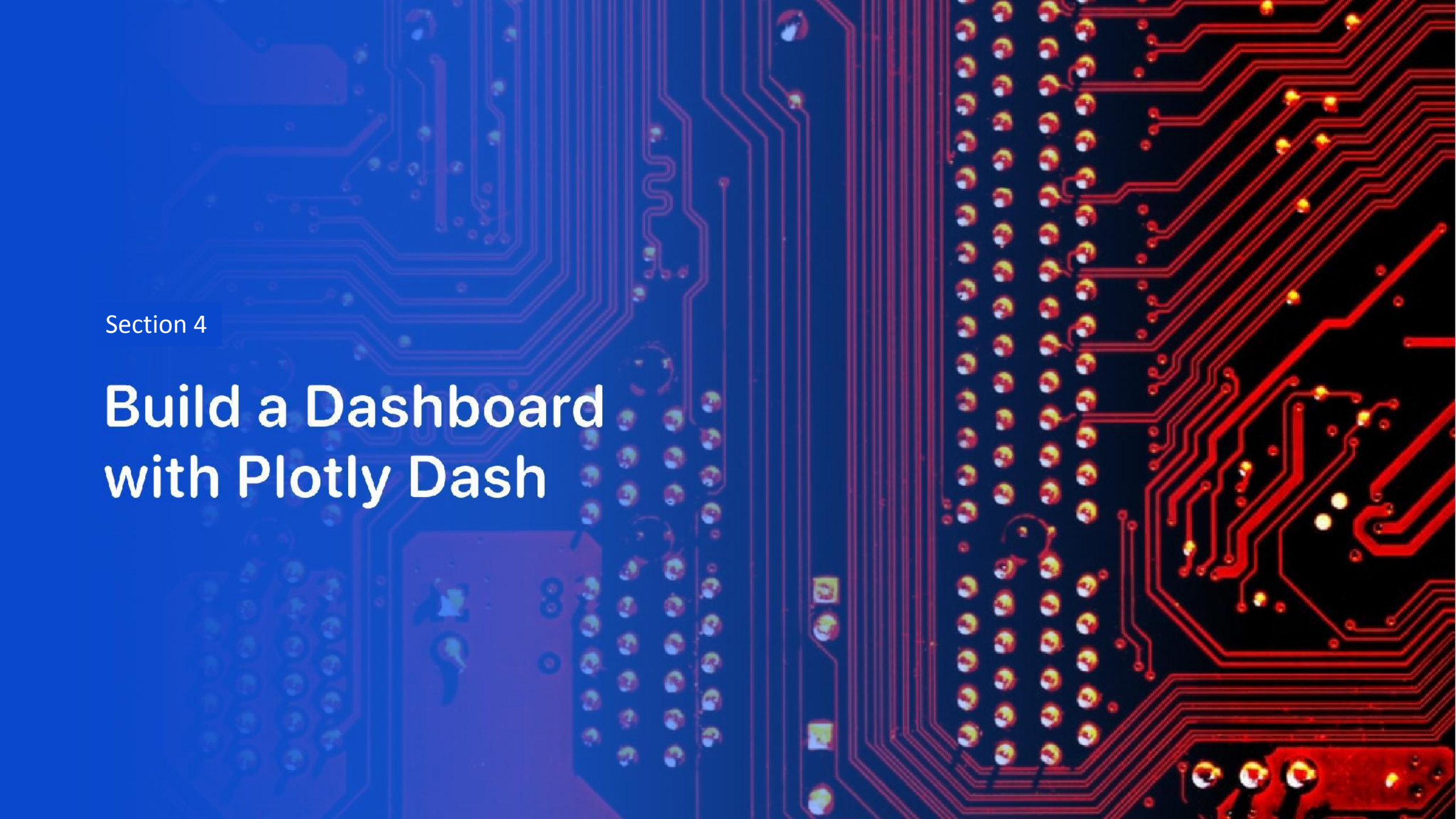


<Folium Map Screenshot 2>



<Folium Map Screenshot 3>

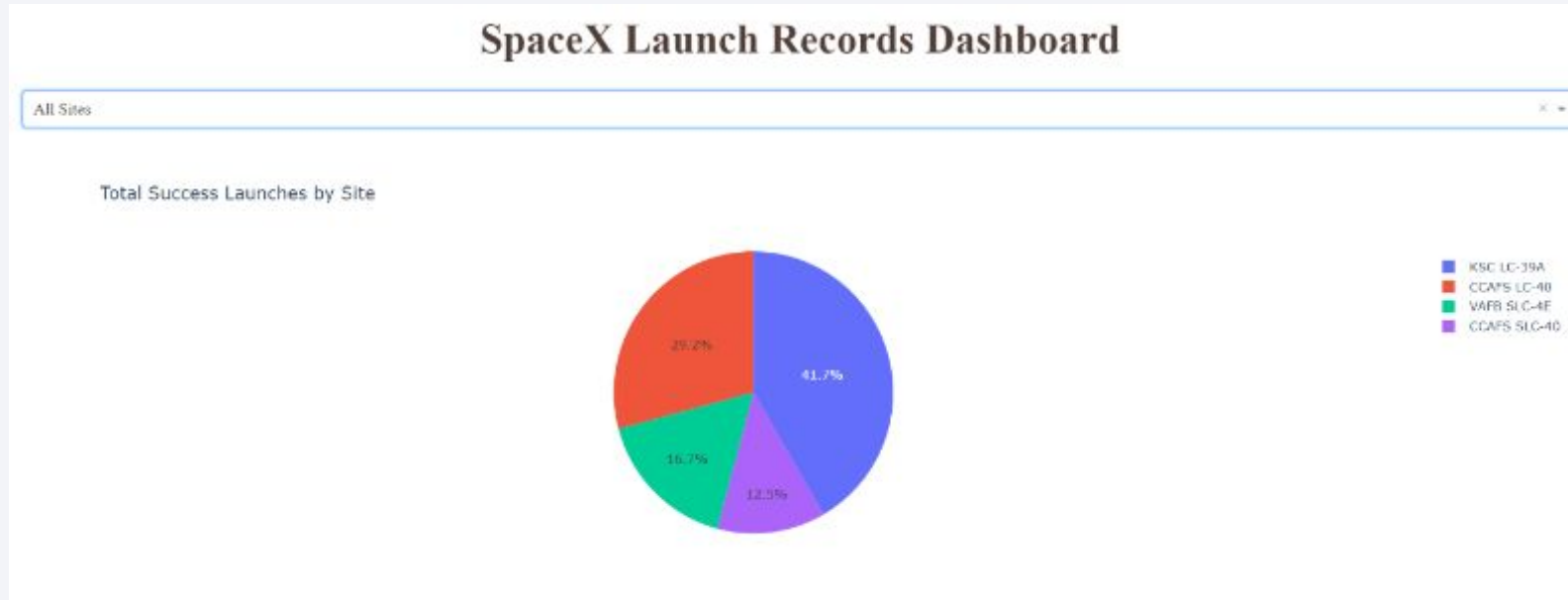
- Replace <Folium map screenshot 3> title with an appropriate title
- Explore the generated folium map and show the screenshot of a selected launch site to its proximities such as railway, highway, coastline, with distance calculated and displayed
- Explain the important elements and findings on the screenshot



Section 4

Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>

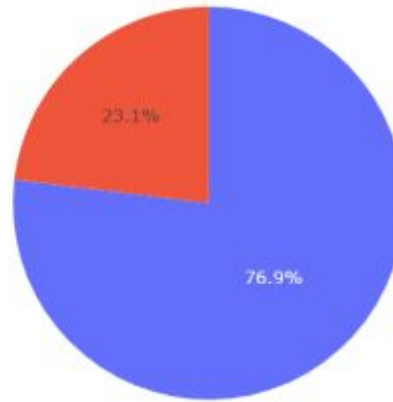


- **KSC LC-39A (Blue):** Holds the highest share of total successful launches at **41.7%**.
- **CCAFS LC-40 (Red):** Represents the second highest share at **29.2%**.
- **VAFB SLC-4E (Green):** Contributes **16.7%** of the total successful launches.
- **CCAFS SLC-40 (Purple):** Accounts for **12.5%** (likely representing a separate naming entry or distinct operational phase in the dataset).

Core Insight: Kennedy Space Center (**KSC LC-39A**) has delivered the largest overall number of successful launches among all facilities in this dataset.

<Dashboard Screenshot 2>

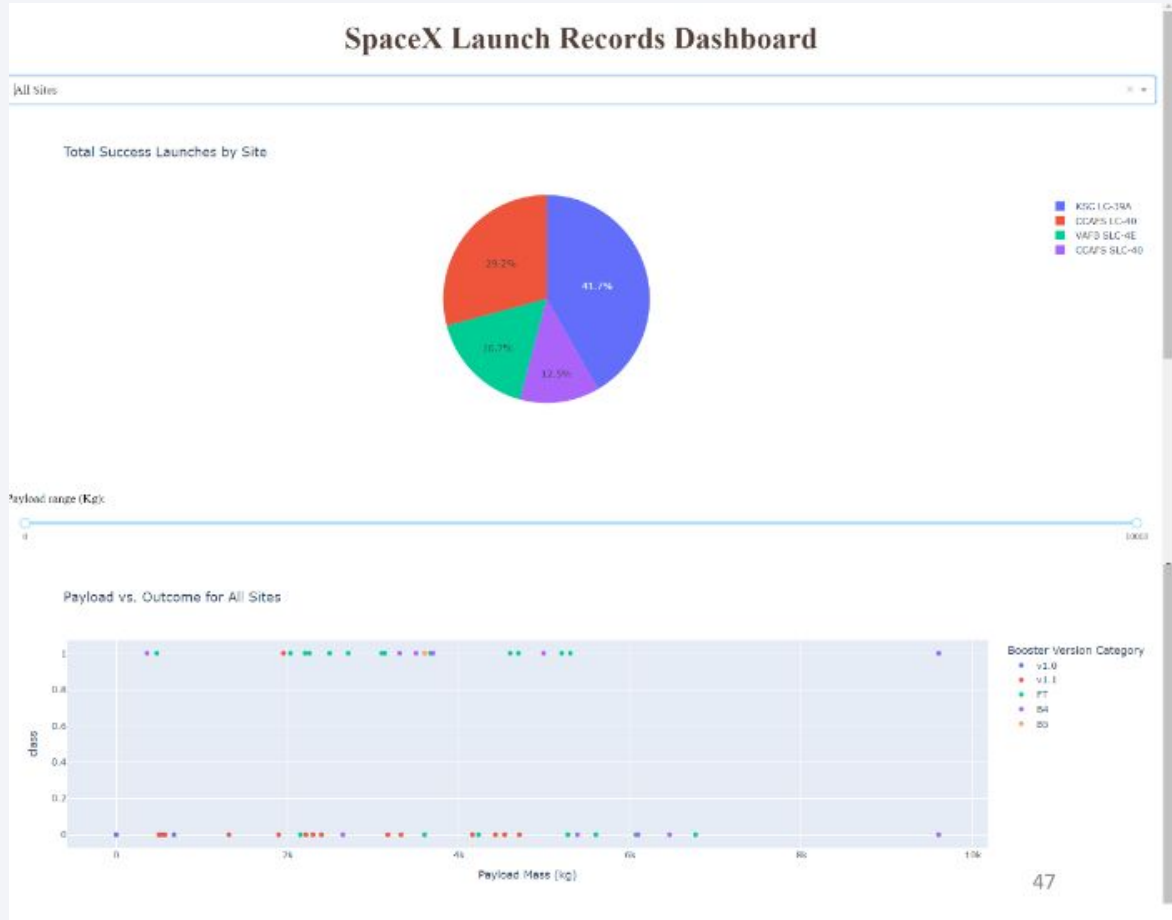
Total Success Launches for site KSC LC-39A



Successes (Blue): Makes up **76.9%** of all launches at this facility.

Failures (Red): Represents **23.1%** of launches.

<Dashboard Screenshot 3>



Top Chart (Site Contribution): KSC LC-39A contributes the highest percentage of total successful launches (**41.7%**), followed by **CCAFS LC-40 (29.2%)**, **VAFB SLC-4E (16.7%)**, and **CCAFS SLC-40 (12.5%)**.
Bottom Chart (Payload vs. Outcome):

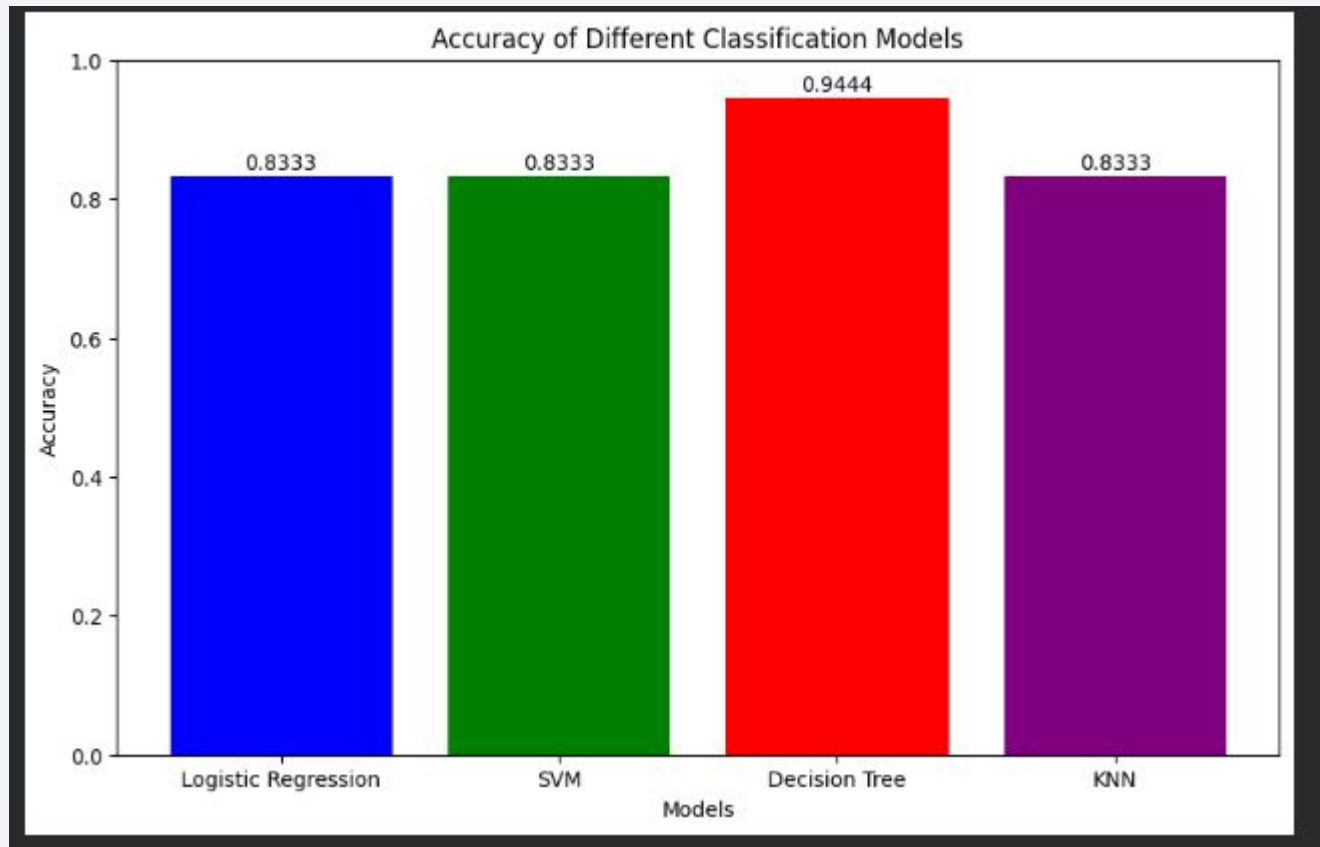
- **Optimal Success Range:** Missions carrying payloads between **2,000 kg** and **4,000 kg** exhibit a dense concentration of successful outcomes (`class = 1`).
- **Booster Reliability Evolution:** Earlier booster versions (`v1.0` in blue and `v1.1` in red) dominate the failure line (`class = 0`), whereas newer booster versions—specifically **FT** (green), **B4** (purple), and **B5** (orange)—account for almost all successful launches across both mid-range and heavy payload weights (> 4,000 kg).



Section 5

Predictive Analysis (Classification)

Classification Accuracy



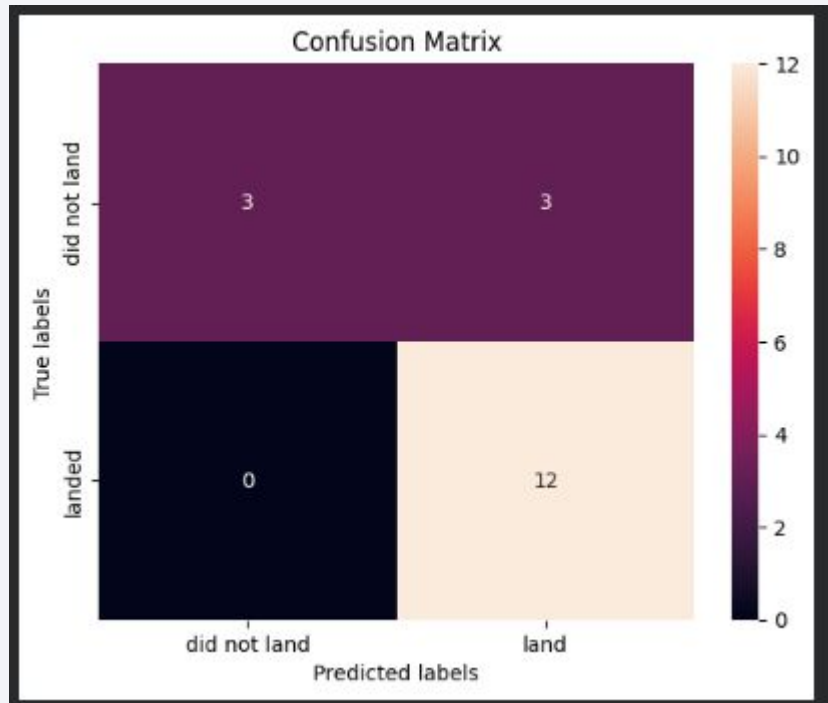
Visualization Focus: Compares the classification accuracy scores across four machine learning algorithms: Logistic Regression, SVM, Decision Tree, and KNN.

Top Performing Model: **Decision Tree** achieves the highest performance with an accuracy of **0.9444** (94.44%).

Tied Classifiers: **Logistic Regression**, **SVM**, and **KNN** all yield an identical accuracy score of **0.8333** (83.33%).

Core Insight: The Decision Tree algorithm significantly outperforms the other baseline models by a margin of roughly 11.1%, making it the optimal classifier for this specific dataset and configuration.

Confusion Matrix



True Negatives (did not land → did not land): 3 samples were correctly identified as failed landings.

False Positives (did not land → land): 3 failed landing instances were incorrectly predicted as successful landings.

False Negatives (land → did not land): 0 samples—the model made zero false negative errors for successful landings.

True Positives (land → land): 12 samples were correctly predicted as successful landings.

Conclusions

- Operational Performance: Launch site selection and booster architecture serve as the strongest indicators of mission outcome. Specifically, CCAFS LC-40 accounted for the highest share of successful launches (43.7%), while the FT (Full Thrust) booster variant demonstrated superior reliability across all payload ranges.
- Payload Mass Dynamics: Data analysis revealed no direct correlation between higher payload masses and lower success rates, proving that booster generation and site-specific operational parameters play a far more critical role in mission completion.
- Predictive Modeling: Among four evaluated classification algorithms (Logistic Regression, SVM, Decision Tree, and KNN), the Decision Tree model yielded the highest performance, achieving superior accuracy in predicting first-stage recovery.

Repository GitHub

<https://github.com/EpsiDelt/Applied-Data-Science-IBM/tree/main>

Thank you!

